



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. ECE-‘A’

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member: Mr.R.Deiva Nayagam

Innovative Practice Description

Unit / Topic: Unit III / Break, Continue, Pass - Differentiation

Course Outcome: CO 3

Topic Learning Outcome: TLO 7, 8

Activity Chosen: Think Pair Share

Justification:

This approach fosters a deeper understanding of these statements by encouraging active thinking, group discussion, and real-world application. Students individually reflect on the applications of break, continue, and pass in Python programming, considering scenarios where each statement can be useful. Within small groups, students discuss their thoughts and share examples of real-world coding situations where break is used to exit loops, continue skips certain iterations, and pass acts as a placeholder. Groups share their insights with the class, promoting a collective understanding of how these statements contribute to code efficiency and readability in various programming contexts.

Time Allotted for the Activity: 20 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 20 minutes. Introduce the concepts of break, continue, and pass in Python, emphasizing their role in controlling loop execution. Explain that the goal is to understand practical applications of these statements in real-world coding scenarios.

Think:

Instructor's Role: Give students individual thinking time to reflect on how break, continue, and pass can be used in Python programming. Encourage considering scenarios where each statement is applicable.

Pair:

Instructor's Role: Form small groups and ask students to pair up within those groups. Instruct pairs to discuss their thoughts on the applications of break, continue, and pass in Python.

Share:

Instructor's Role: Invite each group to share their insights with the class. Facilitate a brief discussion after each presentation to encourage questions and provide additional examples.

Interactive Coding Session:

Instructor's Role: Conduct a live coding session, demonstrating practical examples of using break, continue, and pass in Python. Explain the impact of each statement on the flow of the code, emphasizing their significance in different contexts.

This structured approach allows students to engage in individual reflection, collaborative discussion, and hands-on coding, enhancing their understanding of how to apply "Break, Continue, Pass" in Python programming.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PSO1
CO 3	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	2	1	1	1
Justification for correlation	Introducing Python control flow statements (break, continue, pass) contributes to enhancing students' understanding of fundamental programming concepts.	Students, through individual reflection, can analyze how these control flow statements can be applied to solve programming problems.	Understanding the purpose of break, continue, and pass contributes to the design of efficient and structured Python code.	It provides a foundation for applying fundamental programming concepts to real-world problem-solving.

Image / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- This helped the students to understand the basics of looping statements in Python Programming Language with the help of break, continue and pass statements.
- Students said that the activity is very useful to learn.

❖ *Benefit of the practice:*

- Based on the activity student's felt that it was easy to understand the concept.
- Most of the students were actively participated and enjoyed once they answered correctly in pretest.

❖ *Challenges faced in implementation:*

- Some students only involved in the activity.

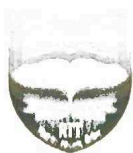
References:

- Amelia, K. R. (2016). Using think-pair-share-strategy to improve vocabulary and reading comprehension achievements of eighth grade students. The Journal of English Literacy Education: The Teaching and Learning of English as a Foreign Language, 3(2), 148-156.

R. Anurag 17/10/23

Signature of Faculty Member

HOD



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Innovative Practice Description

Unit / Topic: Unit I / Illustrative problem: Guess an Integer Number in a Range

Course Outcome: CO 1

Topic Learning Outcome: TLO 1,2,3

Activity Chosen: Role Play

Justification:

- Students are asked to “act out” so they get a better idea of the concepts and theories being discussed. Role play encourages participation among passive learners, adds dynamism to the classroom and promotes the retention of material. Guess an integer in a range is a kind of game where the students are involved to play the game as per the rule.
- **Time Allotted for the Activity:** 20 minutes

Details of the Implementation:

- Begin the class by introducing the concept or theory you want to reinforce through this game.
- Explain the importance of active participation in learning and how role-playing can enhance understanding.
- Divide the students into two to three per group, ensuring that each group has an equal number of participants.
- Define the range of integers that students will be guessing. For example, 1 to 100
- Assign roles within each group, such as a guesser and a moderator.
- The guesser will attempt to guess the integer, while the moderator ensures that the guessing adheres to the rules.
- Integrate short discussion rounds between each guessing attempt, allowing students to share their thought processes and rationale behind their guesses.

- Introduce a scoring system or rewards to make the game more competitive and motivating.
- Reward accurate guesses or creative use of course concepts in explanations.
- After the game, conduct a debriefing session to discuss the experience and link it back to the targeted concepts or theories.
- Encourage students to reflect on how the game enhanced their understanding and retention of the material.
- Conclude the activity with an open discussion, allowing students to share their insights and ask questions.

CO – PO / PSO mapping:

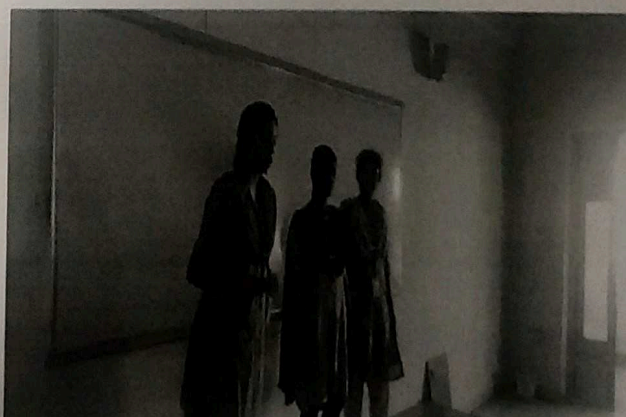
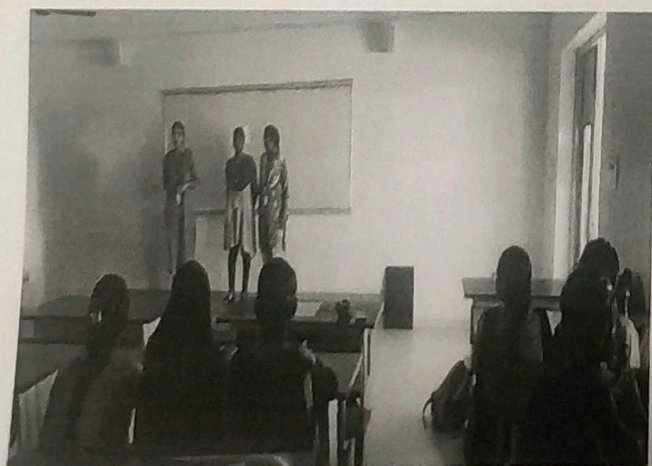
CO	PO1	PO2	PO3	PSO1
CO 1	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	3	2	2	2
Justification for correlation	Apply basics mathematics knowledge to solve the problem	Solutions can be identified, formulated to solve the problem	Solution for the problem can be drawn using flow chart	Helped the students in solving various problems in real world

Images / Screenshot of the practice:



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Through this activity the students felt easy to write algorithm and draw flow chart.

❖ *Benefit of the practice:*

- From this activity, the students can get more clarity in the particular topic.
- Students can able to explain the concepts without any confusion.
- This activity brought out an interest in them and provided a happy & joyful learning environment.
- This practice has made the class more interactive than the conventional practice

Challenges faced in implementation:

- This activity involves only 2 to 3 members.
- Not able to make all the students to involve in this activity

References:

- <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>



Signature of Faculty Member



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